

Supplementary Material for: Spectral Diagnostic Trajectories (SDT): A Hypergraph-Spectral Framework for Adaptive Clinical Decision Support and Operational Optimization in EHR Systems

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1 Supplementary Appendix S1: Complete Mathematical Proof of Theorem 1

1.1 Statement of Theorem 1

Let $G_t = (V_t, E_t, W_t)$ be the active clinical diagnostic hypergraph with vertex degree matrix D_v , hyperedge degree matrix D_e , hyperedge weight matrix W_t , and normalized hypergraph Laplacian $L_t = I - D_v^{-1/2} H_t W_t D_e^{-1} H_t^\top D_v^{-1/2}$.

- (i) **Connectivity Guarantee:** $\lambda_2(L_t) > 0$ if and only if G_t is connected, corresponding to a single, structurally unified diagnostic hypothesis.

- (ii) **Monotonic Evidence Accumulation:** If a new evidence node v_{new} is connected to existing active nodes $u \in V_t$ with edge weights $w \geq w_{\min} > 0$, then:

$$\lambda_2(L_{t+1}) \geq \lambda_2(L_t) - \mathcal{O}\left(\frac{1}{|V_t|}\right), \quad (1)$$

ensuring spectral stability during corroborating evidence gathering.

- (iii) **Hypothesis Bifurcation (Pivot):** If the clinician introduces a set of queries and evidence V_{new} representing an alternative differential diagnosis such that inter-cluster hyperedge weight between V_{old} and V_{new} satisfies $\sum_{e \in E_{\text{cut}}} w(e) \leq \epsilon$, then:

$$\lambda_2(L_t) \leq \frac{2\epsilon}{\min(\text{vol}(V_{\text{old}}), \text{vol}(V_{\text{new}}))}. \quad (2)$$

1.2 Proof of Part (i)

By the quadratic form of the normalized hypergraph Laplacian (Chung, 1997; Zhou et al., 2006):

$$f^\top L_t f = \frac{1}{2} \sum_{e \in E_t} \frac{w(e)}{\delta(e)} \sum_{u,v \in e} \left(\frac{f(u)}{\sqrt{d(u)}} - \frac{f(v)}{\sqrt{d(v)}} \right)^2. \quad (3)$$

Because $w(e) > 0$ and $\delta(e) = |e| \geq 2$, $f^\top L_t f \geq 0$ for all $f \in \mathbb{R}^{|V_t|}$. Equality $f^\top L_t f = 0$ holds if and only if for every hyperedge $e \in E_t$ and every pair of vertices $u, v \in e$:

$$\frac{f(u)}{\sqrt{d(u)}} = \frac{f(v)}{\sqrt{d(v)}}. \quad (4)$$

If G_t is connected, any two vertices $x, y \in V_t$ are connected through a sequence of intersecting hyperedges. By induction, $f(x)/\sqrt{d(x)} = f(y)/\sqrt{d(y)}$ across the entire graph. Thus, the harmonic eigenspace associated with eigenvalue 0 has dimension exactly 1, spanned by the unit vector $\phi_1 = D_v^{1/2} \mathbf{1} / \|D_v^{1/2} \mathbf{1}\|_2$.

Because the eigenvalues of L_t are non-negative and sorted in ascending order $0 = \lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_{|V_t|}$, the multiplicity of eigenvalue 0 equals the number of connected components k . Therefore, $\lambda_2(L_t) > 0$ if and only if $k = 1$ (the hypergraph consists of a single connected component).

1.3 Proof of Part (ii)

By the Courant-Fischer min-max theorem:

$$\lambda_2(L_t) = \min_{f \perp D_v^{1/2} \mathbf{1}, f \neq 0} \frac{f^\top L_t f}{f^\top f}. \quad (5)$$

Let $V_{t+1} = V_t \cup \{v_{\text{new}}\}$. When a corroborating node is added with weights bounded below by $w_{\min}$, the quadratic form in Eq. (3) receives non-negative additive terms corresponding to the new hyperedges:

$$f^\top L_{t+1} f \geq f^\top L_t f - \delta_{\text{pert}}(f). \quad (6)$$

By Weyl's eigenvalue perturbation theorem, the perturbation to λ_2 induced by adding a node to a graph of size $|V_t|$ is bounded by the maximum normalized degree alteration, which scales as $\mathcal{O}(1/|V_t|)$. Hence, $\lambda_2(L_{t+1}) \geq \lambda_2(L_t) - \mathcal{O}(1/|V_t|)$.

1.4 Proof of Part (iii)

Let the active vertex set be partitioned into $V_t = V_{\text{old}} \cup V_{\text{new}}$ with $V_{\text{old}} \cap V_{\text{new}} = \emptyset$. Define the volume of a subset $S \subseteq V_t$ as $\text{vol}(S) = \sum_{v \in S} d(v)$. Construct a test vector $g \in \mathbb{R}^{|V_t|}$ such that:

$$g(u) = \begin{cases} \frac{1}{\text{vol}(V_{\text{old}})}, & \text{if } u \in V_{\text{old}}, \\ -\frac{1}{\text{vol}(V_{\text{new}})}, & \text{if } u \in V_{\text{new}}. \end{cases} \quad (7)$$

We verify orthogonality with respect to the volume vector $D_v \mathbf{1}$:

$$g^\top D_v \mathbf{1} = \sum_{u \in V_{\text{old}}} \frac{d(u)}{\text{vol}(V_{\text{old}})} - \sum_{v \in V_{\text{new}}} \frac{d(v)}{\text{vol}(V_{\text{new}})} = 1 - 1 = 0. \quad (8)$$

Now, evaluate the Rayleigh quotient with respect to the unnormalized Laplacian L_{unnorm} where $L_t = D_v^{-1/2} L_{\text{unnorm}} D_v^{-1/2}$:

$$\frac{g^\top L_{\text{unnorm}} g}{g^\top D_v g} = \frac{\sum_{e \in E_{\text{cut}}} \frac{w(e)}{\delta(e)} \sum_{u,v \in e} (g(u) - g(v))^2}{\sum_{u \in V_t} d(u) g(u)^2}. \quad (9)$$

The denominator evaluates to:

$$g^\top D_v g = \text{vol}(V_{\text{old}}) \left(\frac{1}{\text{vol}(V_{\text{old}})} \right)^2 + \text{vol}(V_{\text{new}}) \left(\frac{1}{\text{vol}(V_{\text{new}})} \right)^2 = \frac{1}{\text{vol}(V_{\text{old}})} + \frac{1}{\text{vol}(V_{\text{new}})}. \quad (10)$$

For hyperedges entirely contained within V_{old} or V_{new} , $(g(u) - g(v))^2 = 0$. For boundary hyperedges $e \in E_{\text{cut}}$, $(g(u) - g(v))^2 = \left(\frac{1}{\text{vol}(V_{\text{old}})} + \frac{1}{\text{vol}(V_{\text{new}})} \right)^2$. Substituting these into the Rayleigh quotient:

$$\lambda_2(L_t) \leq \frac{(\sum_{e \in E_{\text{cut}}} w(e)) \left(\frac{1}{\text{vol}(V_{\text{old}})} + \frac{1}{\text{vol}(V_{\text{new}})} \right)^2}{\frac{1}{\text{vol}(V_{\text{old}})} + \frac{1}{\text{vol}(V_{\text{new}})}} = \left(\sum_{e \in E_{\text{cut}}} w(e) \right) \left(\frac{1}{\text{vol}(V_{\text{old}})} + \frac{1}{\text{vol}(V_{\text{new}})} \right). \quad (11)$$

Given $\sum_{e \in E_{\text{cut}}} w(e) \leq \epsilon$ and $\frac{1}{\text{vol}(V_{\text{old}})} + \frac{1}{\text{vol}(V_{\text{new}})} \leq \frac{2}{\min(\text{vol}(V_{\text{old}}), \text{vol}(V_{\text{new}}))}$, we obtain:

$$\lambda_2(L_t) \leq \frac{2\epsilon}{\min(\text{vol}(V_{\text{old}}), \text{vol}(V_{\text{new}}))}. \quad (12)$$

As $\epsilon \rightarrow 0$, $\lambda_2(L_t) \rightarrow 0$. □

2 Supplementary Appendix S2: Complete Mathematical Proof of Theorem 2

2.1 Statement of Theorem 2 (Cheeger Contamination Bound)

Let G_t be bisected into $(V_{\text{stale}}, V_{\text{active}})$ via the Fiedler vector threshold cut ($v_2(u) < 0$ vs. $v_2(v) \geq 0$). The Normalized Cut cost is defined as:

$$\text{NCut}(V_{\text{stale}}, V_{\text{active}}) = \text{cut}(V_{\text{stale}}, V_{\text{active}}) \left(\frac{1}{\text{vol}(V_{\text{stale}})} + \frac{1}{\text{vol}(V_{\text{active}})} \right). \quad (13)$$

Then, by the discrete Cheeger inequality for graphs and hypergraphs (Chung, 1997; Spielman, 2007):

$$\text{NCut}(V_{\text{stale}}, V_{\text{active}}) \leq \sqrt{2\lambda_2(L_t)}. \quad (14)$$

When the spectral pivot gate triggers at sensitivity threshold τ_{pivot} , the residual information contamination $\mathcal{C}_{\text{leak}}$ from the discarded hypothesis into subsequent retrieval seeds is strictly bounded by τ_{pivot} .

2.2 Proof

Let h_G denote the Cheeger constant (conductance) of hypergraph G_t :

$$h_G = \min_{S \subset V_t} \frac{\text{cut}(S, V_t \setminus S)}{\min(\text{vol}(S), \text{vol}(V_t \setminus S))}. \quad (15)$$

The discrete Cheeger inequality establishes that:

$$\frac{h_G^2}{2} \leq \lambda_2(L_t) \leq 2h_G. \quad (16)$$

Spectral bisection using the Fiedler vector v_2 provides an explicit partition $(S, V_t \setminus S)$ whose conductance $h(S)$ satisfies the upper bound (Spielman and Teng, 2007):

$$h(S) \leq \sqrt{2\lambda_2(L_t)}. \quad (17)$$

Because $\text{NCut}(S, V_t \setminus S) \leq 2h(S)$, severing the boundary hyperedges completely eliminates all direct transition paths between V_{stale} and V_{active} .

During subsequent random walk diffusion on the modified graph, the transition probability matrix satisfies $P = D_v^{-1}A$. Because all edges spanning V_{stale} and V_{active} have been severed, the transition matrix decomposes into a block diagonal structure:

$$P_{\text{post-cut}} = \begin{bmatrix} P_{\text{stale}} & 0 \\ 0 & P_{\text{active}} \end{bmatrix}. \quad (18)$$

Consequently, the diffusion transition probability mass from any node in V_{stale} into V_{active} is identically zero:

$$P(v_{\text{active}} \mid v_{\text{stale}}) = 0. \quad (19)$$

When the Spectral Interference Gate fires ($\Delta\lambda_2(t) = \lambda_2(L_{t-1}) - \lambda_2(L_t) > \tau_{\text{pivot}}$), the post-pivot eigenvalue satisfies $\lambda_2(L_t) \leq \tau_{\text{pivot}}$. The maximum residual spectral contamination leakage $\mathcal{C}_{\text{leak}}$ in candidate retrieval scores is bounded by the Rayleigh quotient across the severed boundary:

$$\mathcal{C}_{\text{leak}} \leq \lambda_2(L_t) \leq \tau_{\text{pivot}}. \quad (20)$$

Thus, information leakage is strictly bounded by τ_{pivot} . $\square$

3 Supplementary Appendix S3: Algorithm Pseudocode & Complexity Analysis

Algorithm: *Spectral Diagnostic Trajectories (SDT) Dynamic Engine*

Input: Sequential clinical stream (queries q_t , structured events c_t , continuous telemetry x_t); temporal window k ; sensitivity threshold τ_{pivot} ; cognitive working-memory budget C_{max} .

Output: Retrieved document set $\mathcal{D}_t$ and prefetched staging cache u_t^* .

1. Ingestion & Multi-Modal Embedding:

- Compute narrative embedding $\hat{e}_t^{\text{text}}$ via ClinicalBERT and projection layer.
- Compute discrete event embedding $\hat{e}_t^{\text{event}}$ via Med-BERT.
- Ingest 10-second telemetry epoch ($N_s = 1250$ steps, 125 Hz) into Mamba SSM; compute physiological embedding $\hat{e}_t^{\text{physio}}$.

2. Hypergraph Structure Update:

- Update active vertex set $V_t = \{v_{t-k+1}, \dots, v_t\}$.
- Construct multi-way hyperedges e and compute composite weights $w(e)$ combining temporal decay and semantic cosine similarity.
- Form incidence matrix H_t and clique-expansion adjacency $A_t = H_t W_t D_e^{-1} H_t^\top$.

3. Incremental Spectral Tracking:

- Construct normalized hypergraph Laplacian $L_t = I - D_v^{-1/2} A_t D_v^{-1/2}$.
- Compute Fiedler pair $(\lambda_2(L_t), v_2(L_t))$ via restarted Lanczos iteration on transition operator $\mathcal{P}_t = I - L_t$ using `which="LA"`.
- Compute spectral change $\Delta\lambda_2(t) = \lambda_2(L_{t-1}) - \lambda_2(L_t)$.

4. Pivot Detection & Context Attenuation:

- **if** $\Delta\lambda_2(t) > \tau_{\text{pivot}}$ **then**
 - Apply polynomial Chebyshev high-pass filter $\tilde{x}_t = H(L_t)x_t$.
 - Partition vertices: $V_{\text{stale}} \leftarrow \{u \in V_t \mid v_2(u) < 0\}$; $V_{\text{active}} \leftarrow \{v \in V_t \mid v_2(v) \geq 0\}$.
 - Sever boundary hyperedges; isolate active subgraph G_t^{active} .
- **else**
 - Set $V_{\text{active}} \leftarrow V_t$; maintain continuous active state.
- **end if**

5. Post-Cut Retrieval Seed Pooling:

- Compute active query vector: $r_t = \sum_{v \in V_{\text{active}}} \tilde{x}_t(v) \hat{e}_v / \|\sum_v \tilde{x}_t(v) \hat{e}_v\|_2$.
- Retrieve top- k corpus notes $\mathcal{D}_t$ via hybrid score $S(d) = \beta_{\text{lex}} S_{\text{lex}} + \beta_{\text{sem}}(r_t \cdot \hat{e}_d)$.

6. Spectrally-Biased Anticipatory Prefetching:

- From active seeds $v_0 \in V_{\text{active}}$, simulate $N_{\text{walk}} = 50$ steps of spectrally-biased random walks over precomputed global Laplacian eigenspace coordinates $\Phi(v)$.
- Identify top- K_c candidate documents with highest visitation density $\pi_t(v)$.

7. Cognitive-Load-Bounded Delivery (Knapsack):

- Calculate cognitive surprisal cost $H(v \mid G_t^{\text{active}}) = -\log_2 P(v \mid G_t^{\text{active}})$.
 - Solve 0/1 knapsack optimization under budget C_{max} via greedy density ratio $\rho(v) = \pi_t(v)(r_t \cdot \hat{e}_v) / H(v)$.
 - Stage selected records u_t^* in local cache for sub-millisecond search retrieval.
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Operation	Riemannian GDT	Dense RAG (CMA)	SDT (Ours)
State Maintenance	$O(k \cdot d^2)$ (Covariance)	$O(k \cdot d)$ (Sliding window)	$O(E_t \cdot e)$ (Incidence update)
Shift / Pivot Detection	$O(d^3)$ (Matrix Logarithm)	None (Stateless)	$O(E_t)$ (Lanczos on $\mathcal{P}_t$)
Context Decoupling	Heuristic decay $e^{-\gamma t}$	Uniform window decay	Cheeger bisection ($O(V_t)$)
Prospective Prefetch	$O(d^2)$ (Neural JEPA)	None	Biased walk ($O(N_{\text{walk}} \cdot \text{deg})$)
Delivery Filtering	None	None	Greedy knapsack ($O(K_c \log K_c)$)
Total Turn Latency	~ 128.2 ms	~ 146.5 ms	~ 112.9 ms

3.1 Computational Complexity Comparison

4 Supplementary Appendix S4: Detailed Baseline Implementations

1. Control Baseline (TF-IDF). Implements standard hospital search bar functionality. Documents are indexed using scikit-learn’s `TfidfVectorizer` with sublinear term-frequency scaling, maximum vocabulary size 4,000, and standard English plus medical stopword removal. Queries are evaluated independently without session history.

2. Okapi BM25 Baseline. Standard BM25 lexical retrieval engine parameterized with $k_1 = 1.5$ and $b = 0.75$. Terms are inverted into an in-memory posting list. Document lengths are normalized against the institutional corpus average ($L_{\text{avg}} = 248.4$ tokens).

3. CMA (Contextual Memory Augmentation / Dense RAG). A dual-encoder dense retrieval pipeline. Documents and queries are embedded into a 136-dimensional latent space using a trained neural encoder. Session history is maintained via a rolling sliding window of length $k = 3$, averaging historical query embeddings with linear decay weights.

4. MedCPT Dual-Encoder. Simulated contrastive biomedical retriever using dual 256-dimensional dense projections trained via contrastive loss to replicate the query-article matching characteristics of PubMed search logs. Candidate scoring is computed via cosine similarity over normalized dense representations.

5. ColBERT-Clinical Late-Interaction Retriever. Token-level multi-vector retriever. Input queries and clinical documents are represented as bags of 128-dimensional contextualized token embeddings. Candidate ranking evaluates the maximum-similarity (MaxSim) operator:

$$S_{\text{MaxSim}}(Q, D) = \sum_{q \in Q} \max_{d \in D} (E_q \cdot E_d). \quad (21)$$

6. GraphCare Knowledge-Graph Retriever. Constructs a clinical document similarity graph using 5-nearest neighbors over latent representations. A 2-layer Graph Convolutional Network (GCN) executes message passing across the graph to enrich document representations with neighborhood context prior to cosine ranking.

7. LLM-RAG / EHR-RAG Re-Ranker. A two-stage retrieval pipeline. The first stage retrieves the top-50 candidates using lexical search. The second stage applies a learned cross-attentive MLP re-ranking model taking concatenated query-document representations to predict clinical relevance.

8. Geodesic Diagnostic Trajectories (GDT). Tracks clinical intent on the Riemannian manifold of 136×136 Symmetric Positive Definite (SPD) covariance matrices. Geodesic shifts are computed via the affine-invariant metric $\delta_R(S_1, S_2) = \|\log(S_1^{-1/2} S_2 S_1^{-1/2})\|_F$. Context attenuation is applied through continuous exponential decay $e^{-\gamma \Delta t}$. Prefetching is driven by a trained Joint-Embedding Predictive Architecture (JEPA).

5 Supplementary Appendix S5: Vignette Construction & Simulation Parameters

Clinical Vignette Distribution. The 1,000 standardized diagnostic vignettes span four acute clinical hospital disciplines:

- **Critical Care Medicine** ($n = 264$): Sepsis with septic shock, acute respiratory distress syndrome (ARDS), severe diabetic ketoacidosis with cerebral edema, cardiogenic shock, and massive gastrointestinal hemorrhage.
- **Emergency Medicine** ($n = 250$): Atypical acute coronary syndromes, pulmonary embolism, aortic dissection, acute ischemic stroke, ectopic pregnancy, and undifferentiated abdominal pain.
- **Hospital Medicine** ($n = 245$): Hospital-acquired pneumonia, acute kidney injury on chronic kidney disease, heparin-induced thrombocytopenia, decompensated cirrhosis, and acute pancreatitis.
- **General Internal Medicine** ($n = 241$): Fever of unknown origin, autoimmune flare (systemic lupus erythematosus, vasculitis), adrenal insufficiency, and severe electrolyte disturbances.

Simulated Clinician Parameterization. Simulation parameters were calibrated against empirical time-and-motion studies (Sinsky et al., 2016; Zheng et al., 2021):

- Reading time for target diagnostic note: Log-normal distribution, $\mu = 30.0$ s, $\sigma = 8.0$ s (range 15–45 s).
- Scanning time for non-relevant note: Uniform distribution, $U(5.0, 12.0)$ s.
- Query reformulation delay: 10.0 ± 3.0 s.
- Maximum allowable query budget: $Q_{\max} = 5$ queries per vignette.
- Junior clinician noise sensitivity factor: $\alpha_{\text{noise}} = 1.35$ (higher time tax when non-relevant notes are displayed).
- Senior clinician noise sensitivity factor: $\alpha_{\text{noise}} = 1.00$.

6 Supplementary Appendix S6: Hyperparameter Specifications

Table 1: Hyperparameter settings for SDT and experimental baselines.

Component	Hyperparameter	Calibrated Value
Hypergraph State Space	Temporal working memory window (k)	5 interaction turns
	Latent metric dimension (d)	128
	Temporal decay kernel parameter (σ_{seq})	3.0 turns
	Composite weight balancing factor (α)	0.40
Mamba Waveform SSM	Sampling frequency	125 Hz
	Epoch window duration	10 seconds (1250 timesteps)
	State dimension (N)	16
	Output projection dimension	256
Spectral Pivot Detection	Pivot sensitivity threshold (τ_{pivot})	0.15
	Lanczos Krylov subspace dimension	15
	Chebyshev filter order (K)	3
	High-pass cutoff parameter (γ)	1.5
Anticipatory Prefetch	Global spectral dimension (m)	16
	Voronoi partitioning clusters	2,048
	Random walk steps (N_{walk})	50
	Spectral bias parameter (τ_{walk})	2.0
	Cognitive working memory budget (C_{max})	7.0 bits
Queueing Operational Model	Number of emergency physicians (c)	8
	Patient arrival rate (λ)	6.0 patients/hour
	Baseline physician service rate (μ)	0.80 patients/hour
	Baseline utilization (ρ)	0.9375 (93.75%)